



LARUS Vario Display

Installation and Operation Manual

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About this document

This manual describes the software for the LARUS Vario Display. This software can be run on various hardware variants. This manual uses the SteFly LARUS Vario Display 57 mm hardware from CoTexx GmbH as a reference.

Software license agreement

This software is released according to the GNU General Public License Version 3. See Appendix A for the full text of the agreement and warranty notice.

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Changelog

v0.1	2025-12-01	First version
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1 General

1.1 Preliminary Remarks

1.1.1 Important Notices

Before using any part of the system, please read and understand this manual. All information in this document is subject to change without notice.

1.1.2 Conventions and Pictograph Definitions

The safety instructions in SteFly operating manuals are the result of risk evaluations and hazard analyses. In this document, the following hazard levels and information are considered:



Pay special attention to critical notes marked with a yellow caution symbol, because non-observance may result in damage or any other critical situation.



A red caution symbol signals that non-observance may result in injuries.



A blue cloud indicates useful information or tips.

1.2 Safety

1.2.1 Safety Precautions



Duty to inform: Each person involved in the installation or operation of LARUS must read and observe the safety-related parts of these operating instructions.

1.2.2 Proper Use

LARUS Vario Display visualises data which are measured and calculated by the LARUS sensor box. LARUS was designed to calculate direction and strength of thermals and wind quickly and reliably. Therefore, the sensor unit combines data from high-precision sensors and GNSS receivers in sophisticated algorithms.

LARUS Vario Display shall be installed in the instrument panel.

LARUS Gliding Sensor Unit is an additional feature to supply glider pilots with accurate information about wind, vertical air movement as well as additional attitude of the aircraft. Its use is limited to day VFR conditions. Security decisions must be made regardless of having installed LARUS.



1.2.3 Improper Use

Improper use will cause all claims for liability and guarantees to be forfeited. Improper use is deemed to be all use for purposes deviating from those mentioned above, especially:

- Using LARUS data in non-VFR conditions or during night is forbidden. LARUS is not certified. Although LARUS provides AHRS data to XCSoar you must not rely on the artificial horizon display.
- Using LARUS data during aerobatics or during flight conditions with high angle of attack (stall) or high g-forces. The algorithm was optimised for normal flight conditions.
- Operating LARUS Vario Display outside the operation conditions defined in technical data section, e.g. input voltage, temperature and humidity.

1.3 Quick Start Manual

LARUS Vario Display may be operated out of the box. Simply perform the following actions:

1. Disassemble both rotary knobs with a 1,5 mm HEX key (scope of delivery)
2. Mount LARUS Vario Display in a 57 mm cutout of the instrument panel with 3 screws. It may be mounted in all four directions.
3. Assemble both rotary knobs
4. Connect the CAN port of LARUS Vario Display and LARUS Sensor Box with a 1:1 patch cable included in scope of delivery.
5. Power on LARUS
6. Check that the satellite symbol on the screen is yellow or green and the current heading is shown.
7. Choose or create a proper polar for your glider.
8. Your LARUS Vario Display is ready to fly!

2 Installation

2.1 Scope of Delivery

- LARUS Vario Display
- Mounting screws
- 1:1 standard RJ45 cable
- Micro SD card with adapter
- D-Sub-9 solder connector and housing
- 1,5 mm HEX key for mounting the rotary knobs

2.2 Design and Function

LARUS Vario Display shows the data measured and calculated by LARUS. LARUS is an advanced variometer with real time wind measurement capabilities. It incorporates state of the art pressure sensors, an advanced IMU and GNSS receivers to gather precise flight information data.

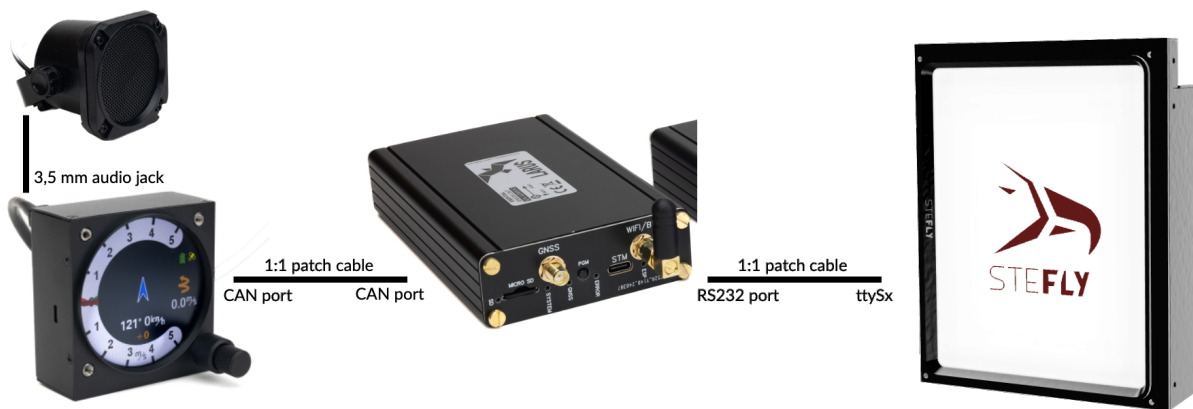
Key features of the display are:

- Round display for 57 mm standard instrument panel cutouts
- Bright and colourful screen
- Lightweight, compact design with black anodised aluminium housing
- Two rotary knobs with push-button function to change settings and enter the menus 1,5 mm HEX key for mounting the rotary knobs

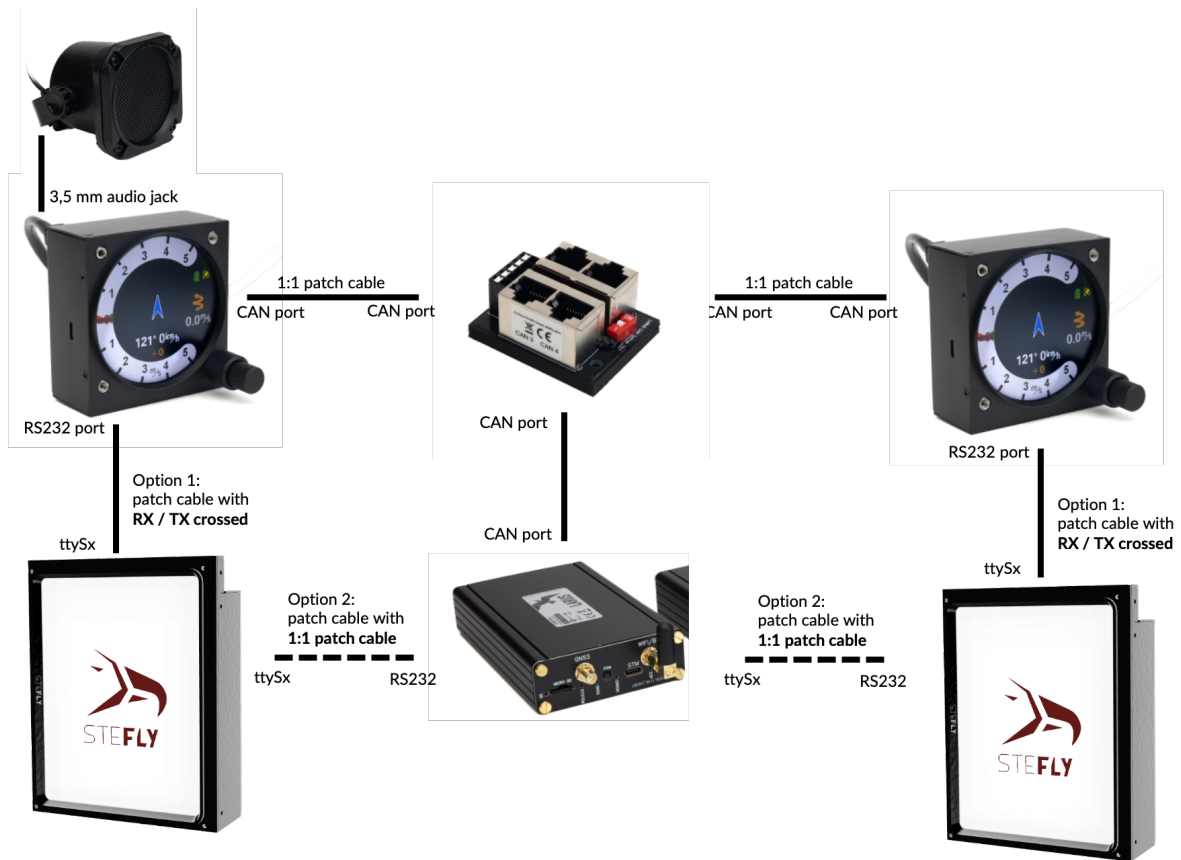
LARUS Vario Display is designed and continuously improved by Prof. Dr. Klaus Schaefer, Maximilian Betz, Winfried Simon, Peter Simon and the SteFly team.

2.3 System Architecture

2.3.1 Single Seater Configuration



2.3.2 Double Seater Configuration



2.4 Connectors and Cabling

2.4.1 Overview of the Connectors



2.4.2 CAN and RS232 Ports

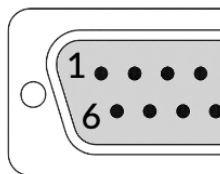


Pin	CAN	RS232
1	GND (internally connected)	GND (internally connected)
2	GND (internally connected)	GND (internally connected)
3	NC	RS232-1-RX
4	CAN Low	RS232-1-TX
5	CAN High	NC
6	NC	NC
7	VCC [9-28V DC]	VCC [9-28V DC]
8	VCC [9-28V DC]	VCC [9-28V DC]

Table 1: Pin Assignment CAN and RS232 RJ45

2.4.3 GPIO / D-SUB 9

Several other switches, sensors and devices can be connected via the d-sub connector. Please note that not all functions have been implemented at date of release of this manual.



The following sketch shows the view into the male connector of the LARUS Vario Display.

For easier pin identification these numbers are also moulded into the female connector (scope of delivery).

After wiring, settings must be made in the LARUS Vario Display.

Pin	CAN	RS232
1	GND	(Ground)
2	DI3 - Gear	Input
3	DI1 - Water Ballast	Input
4	DO2	Output
5	GND	(Ground)
6	DI4 - Speed Breakes	Input
7	DI2 - Spped to fly	Input
8	GND	(Ground)
9	DO1 - Canopy Flasher	Output

Table 2: Pin Assigment GPIO / D-SUB 9

2.4.4 Audio

An audio socket is available for connecting a loudspeaker with a 3.5 mm jack plug. The internal resistance of the speaker must be between 4-8 Ω (max. output of 3 W @ 8 Ω).



A single loudspeaker must not be connected to more than one device.

2.4.5 SD-Card

The device has an SD card slot for firmware updates.



As SD card extensions can damage the LARUS Vario Display, we do not accept any liability for damage resulting from their use.

2.4.6 Cabling

If a serial connection (RS232) is established between SteFly NAV and LARUS Vario Display, the MC value settings, for example, is synchronised/transferred between the devices. In addition, the LARUS Vario Display processes inputs from a Speed to Fly / Vario switch that is connected directly to the SteFly Remote Control Stick. The following steps are required:

- Connect devices with cable:
 - The LARUS Sensor Box CAN-port needs to be connected with 1:1 patch cables with the CAN-port of the LARUS VARIO Display (in double seater configurations using the cable-box)
 - The devices running XCSOAR can be connected between Vario Display RS232 port and SteFly NAV ttySx port with patch cable RX/TX crossed
- Optionally connect switches for automatic adjustment of water ballast and gear warning to the appropriate DSUB-9 pin and any GND pin. For the gear warning you have the option to

use two distinct directly connected switches for the gear and speed brake (recommended) or one cable with the two switches connected in series to the DSUB-9 connector.

- The SteFly Remote Control Stick can be connected in two different ways:
 - if available: with a separate cable to the DSUB-9 pin 7
 - Using the USB-Cable (always available)
- Adjust settings in XCSOar (Version 7.44 or higher) / OpenSoar (7.43 or higher):
 - If the Vario Display is connected directly to the SteFly NAV, select the corresponding ttyS port: Config -> Devices -> select a free line and enter the port, Baud rate 38400, Driver Larus, Sync. to device on
- Adjust settings in LARUS Vario Display settings for the connected optional switches
 - Gear warning
 - water ballast
 - StF Switch configuration (if not automatically switched)
 - * Input Pin: To use an external switch (at the control stick, flaps lever etc), set Vario Control to Input Pin and choose the right setting in StF Pin Config (active: when open/closed)
 - * NMEA for SteFly Remote Control Stick. In addition to the setting in the vario you need to enable install an event definition file in XCSOar/OpenSoar (Config / System / Look / Language, Input / Events – check Expert, click on Events – Download – GLB-XCI-xcremote-XCNAV.xci). Leave XCSOar config and restart XCSOar.

2.5 CAN Termination

LARUS Vario Display and LARUS Sensor Box are connected via CAN. CAN bus networks require termination resistors at each end of the network.

Therefore, all units have integrated a switch to activate the resistor:



CAN termination switch
LARUS Vario Display



CAN termination switch
LARUS Sensor Box



CAN termination switch
CAN Splitter

Please note: All LARUS Sensor Boxes delivered before March 2025 have no CAN termination switch. CAN termination is always on from default.

Description	Display front	Larus Box	CAN Splitter	Display rear
Single Seater	on	on	-	-
Double Seater, Larus box in front panel	off	on	off	on
Double Seater, Larus box in rear panel	on	on	off	off

Table 3: CAN Termination Single / Double Seater


Usually, LARUS Vario Display is powered by LARUS via patch cable between the CAN ports. LARUS needs to be protected by an external fuse (500 mA to max. 3A) like it is common practice for all electric devices in aviation. If LARUS gets its energy from another main instrument (e.g. SteFly NAV via D-Sub connector) please make sure that the main instrument is protected by an external fuse.

2.6 External Fuse

2.7 Installation Location

The following picture shows a typical installation situation of LARUS Vario Display in the instrument panel of a glider.



The display fits in a standard 57 mm cutout and is fastened by 3 M3 screws.



For installation it is required to remove the two rotary knobs with a HEX 1,5mm key.

2.8 Installation Orientation



If you intend to replace an existing instrument with the Lars Vario Display, check the desired installation position before drilling the 7.3 mm hole for the rotary encoder due to the asymmetrical housing to the display.

The display can be mounted in the orientations 0° / 90° / 180° / 270°.

After mounting the display, its orientation needs to be configured in the menu Display Rotation.

2.9 Initial Operation and Function Test

For initial operation please follow these steps:

1. Check that the LARUS Vario Display is connected to LARUS Sensor Box or CAN Splitter according to the drawings in chapter 4.2
2. Power on LARUS
3. Check that LARUS Vario Displays boots and a yellow or green satellite pictogram appears. The vario pointers should perform slight movements around zero position.

2.10 Maintenance

The whole system has no serviceable parts.

To obtain warranty service, please contact SteFly directly.



Opening the housing of LARUS Vario Display will void the warranty!

Problem	Possible Causes	Solution
LARUS Vario Display boots, but satellite pictogram is red and vario pointers are fixed	Connection to LARUS CAN port with Rx / Tx crossed patch cable instead of standard 1:1 patch cable	Change the patch cable and use the one supplied in scope of delivery
LARUS Vario Display boots, but satellite pictogram is red and vario pointers are fixed	Connection to LARUS via RS232 port	Connect the CAN ports instead

Table 4: Troubleshooting

2.11 Troubleshooting

2.12 Firmware Update

The LARUS team is continuously improving the software and releasing updates e.g. on LARUS Vario Display web page.

To update the firmware, put the new .bin-file on the SD card included in scope of delivery and insert it into the SD-card slot of LARUS Vario Display. The device will only install compatible firmware releases greater than the release installed. After powering on, the device is booting within less than a second. If a compatible firmware is detected on the SD Card, the display keeps black for about 3-5 seconds before displaying the a Firmware Upgrade is in process. Afterwards the name of the new firmware version will appear.

If the update is not initialised after powering on the device with an inserted SD card containing a new firmware release, please use a different SD card.

Requirements for the SD card: min. 4 GB SDHC, FAT32 format (not GPT)

3 Operation

3.1 Vario Display

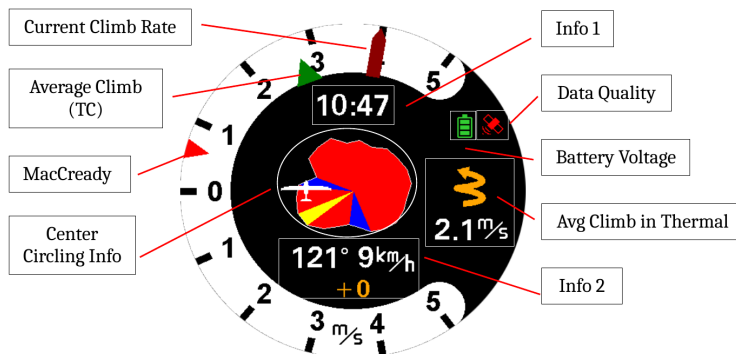
The Central Display Information distinguishes between Circling Mode and Straight Flight Mode. If a turn rate of at least 1°/sec is observed for more than 10 seconds Circling mode is activated. If the turn rate is less than this value for more than 10 seconds it is switched back to Straight Flight Mode.



Switching from Variometer Mode to Speed To Fly Mode is independent from switching the central display information.

For details see chapter Switching from Circling (Vario) to Speed to Fly.

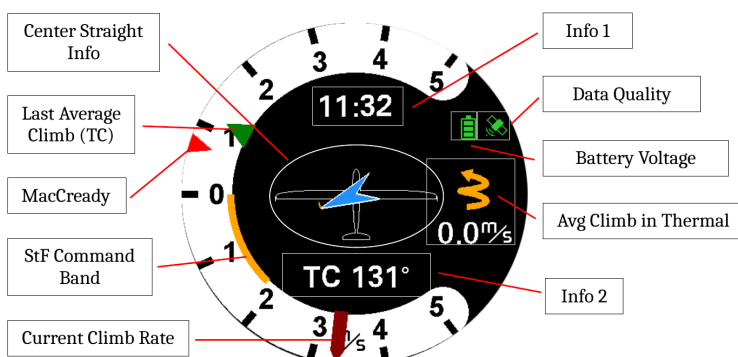
3.1.1 Circling Mode



Content in Circling Mode:

- Info 1: The parameter show is defined in Info 1 Content.
- Data Quality: The colour of the symbol indicates:
 - Green: connection to LARUS Sensor Unit established. Unit has GPS-Fix
 - Yellow: connection to LARUS Sensor Unit established, Unit has no GPS-Fix
 - Red: no connection to LARUS Sensor Unit
- Battery symbol: The colour of the symbol indicates:
 - Green: Voltage of the battery is above Battery Good
 - Yellow: Voltage of the battery is below Battery Good
 - Red: Voltage of the battery is below Battery Low
- Average Climb in Thermal: The average climb rate since starting circling.
- Info 2: The parameter shown is defined in Info 2 Content.
- Current Climb Rate: Time Constant can be changed in Sensor Unit settings Vario TC.
- Average Climb (TC): The average climb rate in the timespan defined in the TC Climb Rate.
- Center Circling Info: Select a Wind or Thermaling Assistance view in Center Circling.

3.1.2 Straight Flight Mode



Displays that differ from Circling Mode:

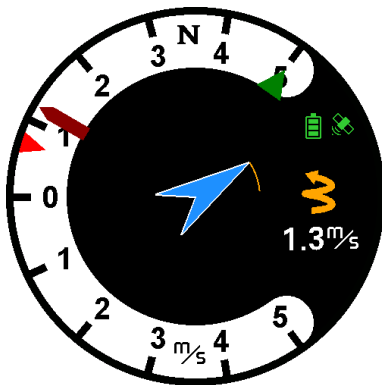
- Center Straight Info: Select one of two available graphical representations

- StF Command Band: The position of the band shows whether you are flying too fast or too slowly. Positive values mean you are flying too fast and negative values mean you are flying too slow. The length of the tape tells you by how much. 1 m/s corresponds to 10 km/h.
- Last Average Climb (TC): The green indicator for the average climb rate is showing the value for the last thermal. The value is not updated until the next switch from speed to fly to vario.

3.1.3 Available Central Displays

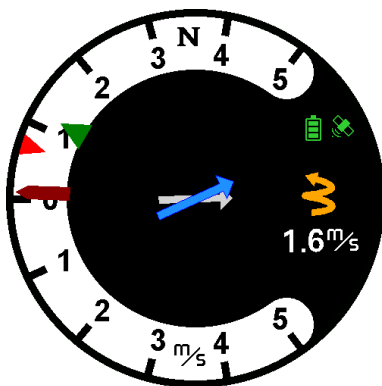
Central Displays in Circling Mode

Single Arrow



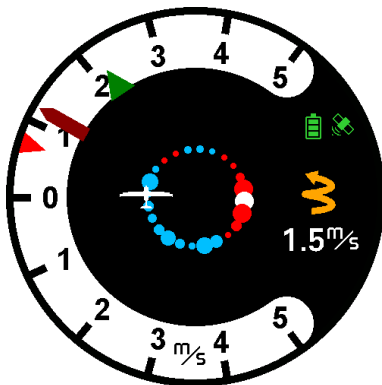
The current wind direction is represented by one central arrow. The arrow size is proportional to the wind speed. Changes of the wind direction to a mid term average direction is show by a trailing vane, changes in wind speed to the mid term average speed is represented by the vane's width. The arrow direction is related to North.

Double Arrow



Current wind direction and speed is represented by the blue arrow, information related to mid term wind is shown by the grey arrow in the background. Arrow sizes depend on the wind speed. The arrow directions are related to North.

Dotted Assistant

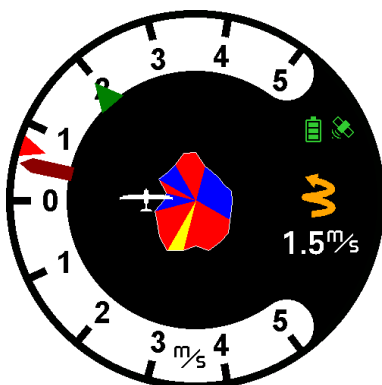


Meaning of the colours of the circle points:

- Blue: climb rate is below the average
- Red: climb rate is about the average
- White/Yellow: climb rate is above the average

The diameter of the dots is proportional to the climb rate.

Spider Assistant



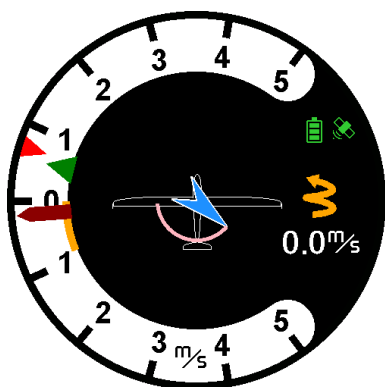
Meaning of the colours of the circle points:

- Blue: climb rate is below the average
- Red: climb rate is about the average
- Yellow: climb rate is above the average

The diameter of the segment is proportional to the climb rate.

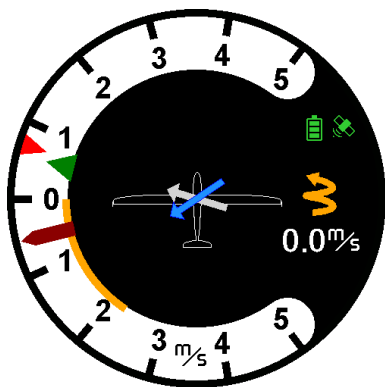
Central Displays in Straight Flight Mode

Single Arrow



The current wind direction is represented by one central arrow. The arrow size is proportional to the wind speed. Changes of the wind direction to a mid term average direction is show by a trailing vane, changes in wind speed to the mid term average speed is represented by the vane's width. The arrow direction is related to the airplanes longitudinal axis. It is advisable to keep the airplane symbol shown (on) in StF Mode in order to support this details.

Double Arrow



Current wind direction and speed is represented by the blue arrow, information related to mid term wind is shown by the grey arrow in the background. Arrow sizes depend on the wind speed. The arrow direction is related to the airplanes longitudinal axis. It is advisable to keep the airplane symbol shown in StF Mode in order to support this details.

3.1.4 Switching from Circling to Speed to Fly

Switching the tone information from Speed to Fly to Vario is depending on your Vario Control configuration:



- Auto: below 1.1 times speed of best glide ratio: Vario mode, above: Speed To Fly mode. If in Vario mode and still circling, Vario mode keeps active even this speed limit is exceeded.
- Input Pin: To use an external switch (at the control stick, flaps lever etc)
- SteFly Remote Control Stick

3.1.5 Acoustic Quiet Zone in Speed to Fly Mode

You can define a speed range in which the acoustic Speed To Fly signal is quiet. Set the desired values in StF Upper/Lower Limit in the LARUS Vario Displays advanced settings.

3.2 Control Elements

The device has 2 rotary knobs and a push button that can be used to make various settings. The following functions are assigned to the button and the two rotary knobs:

- Turn the small/upper rotary knob: Volume control
- Turn the large/lower rotary knob: Set the MacCready
- Short press on the rotary knob: Menu for common settings
- Long press on the rotary knob: Menu for configuration settings of the LARUS Vario Display, configuration and calibration of LARUS Sensor Unit

The following two chapters show the menu structure of the LARUS Vario. Details on the submenu items are documented in the subsequent chapters..

4 Configuration

4.1 Glider Polar

In order to get the right Speed to Fly information you have to select or edit the right polar values for your glider. All glider polars available in XCSOAR/OpenSOAR have been imported into the LARUS Vario. Select the right one in Settings / Polar Settings / Glider.

After selecting the glider you should update the empty mass of your glider.

If you do not find your glider in the list, you can select any glider polar and change the single settings to the values of your gliders polar. You can not change the gliders name.

Every time you enter the glider list in polar settings all values you may have changed before are replaced by the selected gliders hard coded values.

4.2 Calibrating The Larus Sensor Box

Before you start your first flight, you should calibrate the position sensor of the Larus sensor. The individual calibration steps can be carried out using functions in the LARUS vario display.

Calibration is performed in two sections.



4.2.1 Calibration on the Ground (Roll Angle)

Assemble your glider, place it on a flat surface. After taking the individual positions, wait until vibrations in the aircraft can no longer be felt before initiating the measurement.

Do not use a tail dolly in order to fix the vertical axis of your glider during the following operations.

Left Wing

Place the left-hand tip of the wing on the surface. Invoke the function Settings / Sensorbox / Left Wing down.

Right Wing

Place the right-hand tip of the wing on the surface. Invoke the function Settings / Sensorbox / Right Wing down.

Level Wings

Keep the wing horizontal. Invoke the function Setting / Sensorbox / Wings straight

Calculate Roll Angle Sensor Values

For this step it is important to carry out all three steps mentioned before. The order of the steps does not matter, but they must have been completed in full.

Invoke the function Settings / Sensorbox / Calculate Orientation

4.2.2 Calibration During Flight (Pitch Angle)

The pitch angle calibration is performed during flight. It is advisable to perform this step in a flight not disturbed by thermal gusts.

Level your glider out at the speed of the best glide ratio (if you have flaps, with flaps setted for this speed). Invoke Settings / Sensorbox / Straight Flight.

Now you are done. You can check the calibration by swithing to the AHRS Display.



Appendices

A Flight Menu

Flight Menu

- Water Ballast (1)
- Bugs (2)
- Pilot Weight (3)
- Display (4)
- Return

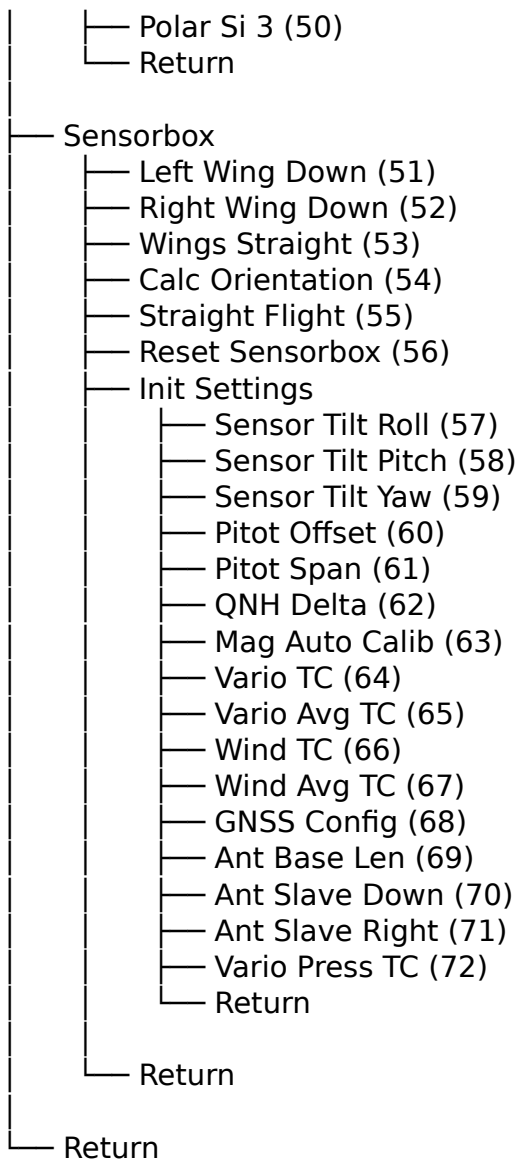
B Settings Menu (Usage Mode Normal)

Settings

- Views
 - Center Circling (5)
 - Center Straight (6)
 - Energy Arrow (7)
 - Display Rotation (8)
 - Glider Symbol (9)
 - Vario
 - Info 1 Content (13)
 - Info 2 Content (14)
 - Info 3 Content (15)
 - Return
 - Speed to Fly
 - Info 1 Content (13)
 - Info 2 Content (14)
 - Info 3 Content (15)
 - Return
 - Units
 - Horizontal Speed (11)
 - Vertical Speed (12)
 - Height (10)
 - Return
 - Return
- Advanced
 - User Profile
 - User Profile (35)
 - Usage Mode (36)
 - Code (73)
 - Config Reset (33)
 - Factory Reset (34)
 - Return



—	Vario
—	— Avg Climb Source (16)
—	— TC Climb Rate (17)
—	— Vario Upper Limit (18)
—	— Vario Lower Limit (19)
—	— Return
—	Speed to Fly
—	— TC Circle Hyst (20)
—	— TC Speed to Fly (21)
—	— Vario Control (22)
—	— StF Pin Config (23)
—	— StF Upper Limit (24)
—	— StF Lower Limit (25)
—	— Return
—	Gear Alarm
—	— Alarm Volume (26)
—	— Gear Alarm Config (27)
—	— Gear Pin Config (28)
—	— Airbrakes Pin Config (29)
—	— Return
—	Drain Control
—	— Drain Pin Config (30)
—	— Lowest Flow (31)
—	— Flow Slope (32)
—	— Return
—	More Settings
—	— Battery Good (38)
—	— Battery Low (39)
—	— Flash Control (40)
—	— Return
—	Center Frequency (37)
—	— Return
—	Polar Settings
—	— Glider (41)
—	— Empty Mass (42)
—	— Max Ballast (43)
—	— Reference Weight (44)
—	— Polar V 1 (45)
—	— Polar V 2 (46)
—	— Polar V 3 (47)
—	— Polar Si 1 (48)
—	— Polar Si 2 (49)



C Settings Menu (Usage Mode Club)

